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INKJET PRINTING APPARATUS

Abstract

In this inkjet printing apparatus, an ink separation region is set in a partial region including a heater in a feedback pipe that connects an internal collecting tank and a supply tank on an ink circulation path. The ink separation region is a region in which ink in the circulation path is separated from ink in the other regions in the feedback pipe when ink circulation stops. A control unit selectively performs a normal resumption mode and a post-drainage resumption mode in accordance with a temperature of ink in the ink separation region before resuming ink circulation in the circulation path. In the post-drainage resumption mode, after at least a part of ink in the ink separation region is drained to an external collecting tank, the ink separation region and the other regions in the feedback pipe are caused to communicate with each other.

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Background/Summary

RELATED APPLICATIONS
[0001] This application claims the benefit of Japanese application No. 2024-035467, filed on 8 Mar. 2024, the disclosure of which is incorporated by reference herein.
BACKGROUND OF THE INVENTION
Field of the Invention
[0002] The present invention relates to an inkjet printing apparatus that discharges ink onto a printing medium such as paper to perform printing.
Description of the Background Art
[0003] Conventionally, an inkjet printing apparatus includes an ink circulation path through which ink is supplied to a discharge head configured to discharge ink onto a printing medium, and ink being left undischarged in the discharge head is collected and again supplied to the discharge head, in some cases. An inkjet printing apparatus including such an ink circulation path as mentioned above is described in, for example, Japanese Patent Application Laid-Open No. 2012-006261.
[0004] An image recorder (1) in Japanese Patent Application Laid-Open No. 2012-006261 includes an ink circulation unit (3a). The ink circulation unit (3a) includes a first ink circulation path (14) and a second ink circulation path (15), and thus, two kinds of ink are circulated. Further, the first ink circulation path (14) and the second ink circulation path (15) are connected to an image recording unit (4). Ink is circulated through each of the first ink circulation path (14) and the second ink circulation path (15), and thus ink is supplied to each nozzle row (11) of the image recording unit (4) (paragraph [0027]). Meanwhile, a heat exchanger (23a) is provided across the first ink circulation path (14) and the second ink circulation path (15) (paragraph [0028] and FIG. 2).
[0005] Here, for example, consider a case in which an image is recorded while ink is circulated through only the first circulation path (14) to which ink of K color is charged. In this case, when ink is discharged from the nozzle row (11), driving heat is generated in the image recording unit (4), and a temperature of ink being delivered through the first ink circulation path (14) is gradually increased. At that time, in the heat exchanger (23), whereas ink therein is cooled by heat dissipation using a heat-dissipating fin (65), the cooling performance is gradually degraded (paragraphs [0041] and [0042]). Then, a control unit (2) starts circulating ink through the second ink circulation path (15) that is not used in image recording (paragraph [0043]). In this regard, a temperature of ink in the second circulation path (15) is low because the second circulation path (15) has not been used in image recording. The ink at a low temperature is delivered to the heat exchanger (23), and thus, heat can be absorbed and the temperature can be lowered (paragraph [0044]).
[0006] However, if the image recording apparatus (1) and the ink circulation unit (3a) stop due to a reason such as a power failure, delivery of ink through the second ink circulation path (15) to the heat exchanger (23) for ink also stops. In such a case, the temperature of ink in the first circulation path (14) is kept high and is likely to be changed in quality or deteriorated. To directly use ink having been changed in quality or deteriorated for image recording would cause a fear that ink might not be satisfactorily discharged from the nozzle row (11) or a desired color gamut might not be obtained. For this reason, in recovering after a stop of the image recording apparatus (1) and the ink circulation unit (3a), it is required to perform waste liquid treatment on ink having been changed in quality or deteriorated. However, to perform waste liquid treatment on all of ink in the first ink circulation path (14) would incur much loss, probably reducing workability.

SUMMARY OF THE INVENTION

[0007] It is an object of the present invention to provide a technology that enables drainage of ink that is likely to have been changed in quality or deteriorated due to overheating during a stop of ink circulation, to the outside, in recovery afterward, and further, enables reduction of an amount of ink to be drained at that time.

[0008] To solve the above-described problem, the first invention of the present application is directed to an inkjet printing apparatus that discharges ink onto a printing medium to perform printing, and includes an ink circulation path, an external collecting tank, a circulation pump, a heater, a control unit, and an ink separation region. The circulation path includes a discharge head configured to discharge ink, a supply tank in which ink supplied to the discharge head is stored, an internal collecting tank in which ink collected from the discharge head is stored, and a feedback pipe connecting the internal collecting tank and the supply tank. In the external collecting tank, ink drained to outside of the circulation path is collected. The circulation pump is interposed in the feedback pipe and is configured to deliver ink from the internal collecting tank to the supply tank via the feedback pipe. The heater is interposed between the circulation pump and the supply tank in the feedback pipe and is configured to heat ink flowing from the internal collecting tank to the supply tank. The control unit is configured to control the discharge head, the circulation pump, and the heater, and detect presence or absence of ink circulation in the circulation path. The ink separation region is set in a partial region including the heater in the feedback pipe. In the ink separation region, ink in the circulation path is separated from ink in the other regions in the feedback pipe when the ink circulation in the circulation path stops. The control unit selectively performs a normal resumption mode and a post-drainage resumption mode before resuming the ink circulation in the circulation path in accordance with a temperature of the ink in the ink separation region. In the normal resumption mode, the ink separation region and the other regions in the feedback pipe are caused to communicate with each other immediately. In the post-drainage resumption mode, the ink separation region and the other regions in the feedback pipe are caused to communicate with each other after at least a part of ink in the ink separation region is drained to the external collecting tank.

[0009] The second invention of the present application is directed to the inkjet printing apparatus according to the first invention, further including an on-off valve and a switch valve in the ink separation region. The on-off valve is interposed between the circulation pump and the heater in the feedback pipe and is configured to allow or interrupt ink flow from the internal collecting tank to the supply tank. The switch valve is interposed between the heater and the supply tank in the feedback pipe and is configured to perform switching between a state in which the heater and the supply tank communicate with each other and a state in which the heater and the supply tank are blocked from communicating with each other, and in the latter state, an inner space of the circulation path and the external collecting tank communicate with each other. The control unit is configured to further control the on-off valve and the switch valve. Under the control of the control unit, the on-off valve is closed, and the ink separation region and the external collecting tank are caused to communicate with each other at the switch valve, so that the ink in the ink separation region is separated from the ink in the other regions in the feedback pipe.

[0010] The third invention of the present application is directed to the inkjet printing apparatus according to the second invention, wherein, in the post-drainage resumption mode, under the control of the control unit, the on-off valve is opened, and the circulation pump is driven while the ink separation region and the external collecting tank communicate with each other at the switch valve, so that at least the part of the ink in the ink separation region is drained to the external collecting tank.

[0011] The fourth invention of the present application is directed to the inkjet printing apparatus according to any of the first to third inventions, wherein the control unit determines whether the temperature of the ink in the ink separation region is lower than a predetermined threshold value, or is equal to or higher than the predetermined threshold value, performs the normal resumption mode when it is determined that the temperature of the ink in the ink separation region is lower than the predetermined threshold value, and performs the post-drainage resumption mode when it is determined that the temperature of the ink in the ink separation region is equal to or higher than the predetermined threshold value.

[0012] The fifth invention of the present application is directed to the inkjet printing apparatus according to the fourth invention, wherein the control unit determines whether the temperature of the ink in the ink separation region is lower than, or equal to or higher than, the predetermined threshold value on the basis of a duty value of the heater for a first predetermined time period before it is detected that the ink circulation in the circulation path has stopped.

[0013] The sixth invention of the present application is directed to the inject printing apparatus according to the fourth invention, further including a temperature sensor that is interposed between the heater and the switch valve in the feedback pipe and is configured to detect a temperature of ink in the feedback pipe. The control unit determines whether the temperature of the ink in the ink separation region is lower than, or equal to or higher than, the predetermined threshold value on the basis of a result of detection provided from the temperature sensor.

[0014] The seventh invention of the present application is directed to the inkjet printing apparatus according to any of the first to sixth inventions, wherein, in performing the post-drainage resumption mode, the control unit stops driving the heater at least until the ink separation region and the other regions in the feedback pipe communicate with each other.

[0015] The eighth invention of the present application is directed to the inkjet printing apparatus according to the second invention, further including a first heater and a second heater that are interposed in the feedback pipe while being arranged adjacently to each other along the circulation path, as the heater. The second heater is positioned on a downstream side of the first heater in the circulation path and has a duty value greater than a duty value of the first heater. Before resuming the ink circulation in the circulation path, the control unit determines whether a temperature of ink in the first heater is lower than, or equal to or higher than, a first predetermined threshold value, and whether a temperature of ink in the second heater is lower than, or equal to or higher than, a second predetermined threshold value. When it is determined that the temperature of the ink in the first heater is lower than the first predetermined threshold value and the temperature of the ink in the second heater is equal to or higher than the second predetermined threshold value, the control unit performs the post-drainage resumption mode in which the inner space of the circulation path and the external collecting tank are caused to communicate with each other at the switch vale, the circulation pump is driven for a specific time period, the ink in the ink separation region is drained to the external collecting tank, and then, the ink separation region and the other regions in the feedback pipe are caused to communicate with each other at the switch valve. The specific time period is obtained by division of a specific capacity calculated from an ink-passable capacity of the second heater, by a pumping amount of the circulation pump per unit time period.

[0016] According to the first to eighth inventions of the present application, ink that is likely to have been changed in quality or deteriorated due to overheating during a stop of ink circulation can be drained to the outside in accordance with a temperature of ink, in recovery afterward. Further, a smaller amount of ink than all ink in the circulation path can be drained, and hence, an amount of ink to be drained can be reduced.

[0017] Especially, according to the second invention of the present application, ink in the ink separation region is separated from ink in the other regions in the feedback pipe using the on-off valve and the switch valve, which ensures separation of ink.

[0018] Especially, according to the third invention of the present application, in the post-drainage resumption mode, ink is drained to the external collecting tank by driving of the circulation pump, which ensures drainage of ink.

[0019] Especially, according to the fifth invention of the present application, ink can be drained in

accordance with a result of determination as to whether the ink temperature is lower than, or equal to or higher than, the threshold value, with consideration of a heating value of the heater.
[0020] Especially, according to the sixth invention of the present application, the temperature of the ink in the ink separation region is detected using the temperature sensor, which enables more accurate detection of the ink temperature.
[0021] Especially, according to the seventh invention of the present application, overheating of ink can be further suppressed.
[0022] Especially, according to the eighth invention of the present application, ink that is likely to have been changed in quality or deteriorated due to overheating by the second heater can be drained to the outside in accordance with the temperature of the ink. Further, it is possible to narrow ink to be drained at that time, down to ink that is likely to have been changed in quality or deteriorated due to overheating by the second heater.
[0023] These and other objects, features, aspects and advantages of the present invention will become more apparent from the following detailed description of the present invention when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0024] FIG. **1** is a view conceptually showing a configuration of an inkjet printing apparatus;
[0025] FIG. **2** is a view conceptually showing configurations of an ink supply unit and a discharge head;
[0026] FIG. **3** is a block diagram showing connection between a control unit and each component of the inkjet printing apparatus;
[0027] FIG. **4** is a flowchart showing a flow including conveyance of a continuous paper, printing on the continuous paper, ink circulation, and ink drainage to the outside;
[0028] FIG. **5** is a view showing a relationship between a temperature of ink passing through a heater and a duty value of the heater, plotted against the temperature; and
[0029] FIG. **6** is a flowchart showing a procedure for performing a post-recovery process.
DESCRIPTION OF THE PREFERRED EMBODIMENTS
[0030] Hereinafter, a preferred embodiment of the present invention will be described with reference to the drawings. Note that components described in the preferred embodiment are mere examples and are not intended to limit the scope of the present invention to those only. In the drawings, for the purpose of easier understanding, the dimensions or the number of respective components are overstated or understated in some portions of illustration, as necessary.
<1. Configuration of Inkjet Printing Apparatus>
[0031] FIG. **1** is a view conceptually showing a configuration of an inkjet printing apparatus **1** according to one preferred embodiment of the present invention. The inkjet printing apparatus **1** is an inkjet printing machine that discharges droplets of water-based ink, from a plurality of discharge heads **35**, onto continuous paper **10** being in a shape of a long strip, while conveying the continuous paper **10**, to record characters or images on a surface of the continuous paper **10**. Note that the continuous paper **10** in a shape of a long strip is just one example of a printing medium. The printing medium may be a cut sheet, a plastic film, a cardboard, metal foil, a glass material, or the like. In other words, the inkjet printing apparatus **1** may be any apparatus that can discharge ink onto a printing medium to perform printing. As shown in FIG. **1**, the inkjet printing apparatus **1** includes a conveyor unit **2**, a printing unit **3**, and a control unit **9**.
[0032] The conveyor unit **2** is a mechanism configured to convey the continuous paper **10** along a predetermined conveying path in a conveying direction extending along a length direction of the continuous paper **10**. The continuous paper **10** is stretched over a plurality of conveyor rollers **12**.

The continuous paper **10** is conveyed along a conveying path formed by the plurality of conveyor rollers **12**. Each of the conveyor rollers **12** rotates about an axis extending in a direction perpendicular to the conveying direction, to thereby guide the continuous paper **10** to the downstream side in the conveying path. Further, the continuous paper **10** is under tension in the conveying direction. This reduces slack or wrinkles in the continuous paper **10** during conveying.

[0033] The printing unit **3** includes a plurality of discharge heads **35** and a plurality of ink supply units **4**. In the present preferred embodiment, the printing unit **3** includes four discharge heads **35** and four ink supply units **4**. The four discharge heads **35** have substantially the same configuration with each other. Further, the four ink supply units **4** have substantially the same configuration with each other.

[0034] The four discharge heads **35** are arranged while being spaced from each other along the conveying direction. Each of the four discharge heads **35** discharges ink droplets onto a surface (upper surface) of the continuous paper **10** from nozzles **83** (refer to FIG. **2** described later). In the present preferred embodiment, the four discharge heads **35** discharge ink of different colors, respectively, to thereby each record a monochromatic image on the surface (upper surface) of the continuous paper **10**. For example, the four discharge heads **35** discharge cyan ink, magenta ink, yellow ink, and black ink, respectively. Then, the four monochromatic images are superimposed, so that a multicolor image is formed on the upper surface of the continuous paper **10**.

[0035] FIG. **2** is a view conceptually showing a configuration of one ink supply unit **4** and a configuration of one discharge head **35**. In the present preferred embodiment, each of the discharge heads **35** includes a plurality of heads **80**. In the present preferred embodiment, each of the discharge heads **35** includes five heads **80**. The five heads **80** have substantially the same configuration with each other. Hence, in FIG. **2**, only one of the five heads **80** is shown in detail, and the other four heads **80** are shown in a simplified manner. As shown in FIG. **2**, each of the five heads **80** includes a casing **81**, an internal tank **82**, and a plurality of nozzles **83**.

[0036] The casing **81** forms an outer frame of the head **80**. The internal tank **82** is provided in the casing **81**, and ink can be temporarily stored therein. The plurality of nozzles **83** are arranged while being equally spaced from each other along the conveying direction and a width direction of the continuous paper **10** in a lower portion of the casing **81**. Each of the plurality of nozzles **83** communicates with the internal tank **82**. Further, each of the plurality of nozzles **83** includes a plurality of piezoelectric elements **831** serving as pressure generation elements, an ink chamber **832**, and a discharge port **830**. The ink chamber **832** communicates with the internal tank **82**.

[0037] During discharge of ink, ink flows down from the internal tank **82** to the ink chamber **832**. Then, under the control of the piezoelectric elements **831**, ink in the ink chamber **832** is pressurized, and thus is discharged in the form of liquid droplets from the discharge port **830**. Alternatively, the nozzle **83** may be a thermal nozzle in which ink in the ink chamber **832** is heated to generate bubbles and thus is pressurized.

[0038] Next, the ink supply unit **4** is described. The ink supply unit **4** is a device configured to supply ink to the discharge heads **35** while circulating a part of ink. As described above, the inkjet printing apparatus **1** of the present preferred embodiment includes four ink supply units **4**. The four ink supply units **4** have substantially the same configuration with each other, and hence only a configuration of one ink supply unit **4** is described below.

[0039] As shown in FIG. **2**, each of the ink supply units **4** includes a supply tank **51**, an internal collecting tank **52**, an external collecting tank **53**, a supply-side manifold **61**, a plurality of supply-side narrow pipes **62**, a plurality of collecting-side narrow pipes **63**, a collecting-side manifold **64**, a feedback pipe **65**, a drainage pipe **66**, a circulation pump **71**, a plurality of supply-side on-off valves **73**, a plurality of head outlet-side on-off valves **74**, a feedback-side on-off valve **75**, a first heater **76**, a second heater **77**, a switch valve **78**, a first temperature sensor **84**, a second temperature sensor **85**, a third temperature sensor **86**, a filter **87**, and a deaeration unit **88**. In the present preferred embodiment, each of the ink supply units **4** includes five supply-side narrow pipes **62**, five collecting-side narrow pipes **63**, five supply-side on-off valves **73**, and five head outlet-side on-off valves **74**.

[0040] The supply tank **51** is a container for temporally storing ink to be supplied to the discharge heads **35**. In the supply tank **51**, an internal chamber **510** in which ink can be temporally stored is provided. Meanwhile, in the supply tank **51**, a liquid-level sensor for detecting a liquid level of ink stored in the internal chamber **510** of the supply tank **51** may be provided.

[0041] The supply-side manifold **61** and the five supply-side narrow pipes **62** are pipes connecting the supply tank **51** and the five heads **80** included in one discharge head **35**. The supply-side manifold **61** is a wide pipe having an upstream end that is connected so as to communicate with the internal chamber **510** of the supply tank **51**. Each of the five supply-side narrow pipes **62** is a narrow pipe branching from the supply-side manifold **61**. Each of the five supply-side narrow pipes **62** has an upstream end communicating with an internal passage of the supply-side manifold **61**, and has a downstream end that is connected so as to communicate with the internal tank **82** of one head **80**.

[0042] Further, in the present preferred embodiment, the supply-side on-off valve **73** is interposed in each of the supply-side narrow pipes **62**. For the supply-side on-off valve **73**, for example, a solenoid valve that is opened and closed under the control of the control unit **9** is used. Alternatively, for the supply-side on-off valve **73**, an on-off valve that is manually opened and closed may be used. While the supply-side on-off valve **73** is closed, an internal passage of the supply-side narrow pipe **62** is blocked from communicating. That is, while the supply-side on-off valve **73** is closed, ink flow from the supply tank **51** to the head **80** is interrupted. Meanwhile, while the supply-side on-off valve **73** is opened, the internal passage of the supply-side narrow pipe **62** is allowed to communicate. Note that the supply-side on-off valve **73** is not necessarily required to be provided. Further, a filter or the like may be further interposed in the supply-side manifold **61** or each of the five supply-side narrow pipes **62**.

[0043] The five collecting-side narrow pipes **63** and the collecting-side manifold **64** are pipes connecting the five heads **80** included in one discharge head **35** and the internal collecting tank **52**. Each of the five collecting-side narrow pipes **63** is a narrow pipe branching from the collecting-side manifold **64**. Each of the five collecting-side narrow pipes **63** has an upstream end that is connected so as to communicate with the internal tank **82** of one head **80**, and has a downstream end that is connected so as to communicate with an internal passage of the collecting-side manifold **64**. The collecting-side manifold **64** is a wide pipe having a downstream end that is connected so as to communicate with the internal chamber **520** of the internal collecting tank **52** described later.

[0044] Further, in the present preferred embodiment, the head outlet-side on-off valve **74** is interposed in each of the collecting-side narrow pipes **63**. For the head outlet-side on-off valve **74**, for example, a solenoid valve that is opened and closed under the control of the control unit **9** is used. Alternatively, for the head outlet-side on-off valve **74**, an on-off valve that is manually opened and closed may be used. While the head outlet-side on-off valve **74** is closed, an internal passage of the collecting-side narrow pipe **63** is blocked from communicating. That is, while the head outlet-side on-off valve **74** is closed, ink flow from the head **80** to the internal collecting tank **52** is interrupted. Meanwhile, while the head outlet-side on-off valve **74** is opened, the internal passage of the collecting-side narrow pipe **63** is allowed to communicate. Note that the head outlet-side on-off valve **74** is not necessarily required to be provided. Further, a filter or the like may be further interposed in each of the five collecting-side narrow pipes **63** or the collecting-side manifold **64**.

[0045] The internal collecting tank **52** is a container for temporally storing ink collected from the discharge heads **35**. In the internal collecting tank **52**, the internal chamber **520** in which ink can be temporally stored is provided. Meanwhile, in the internal collecting tank **52**, a liquid-level sensor for detecting a liquid level of ink stored in the internal chamber **520** of the internal collecting tank **52** may be provided.

[0046] Further, as shown in FIG. **2**, the supply tank **51** is connected to a pressurization mechanism **515**. The pressurization mechanism **515** pressurizes the inside of the supply tank **51**, to regulate a pressure of the internal chamber **510** of the supply tank **51** to a positive pressure. That is, the pressurization mechanism **515** pressurizes the inside of the supply tank **51**, to regulate a pressure of the internal chamber **510** of the supply tank **51** to a pressure higher than the atmospheric pressure. The pressurization mechanism **515** includes, for example, a compressor, a pressurization buffer tank, a pressure regulation mechanism (regulator), and the like. Meanwhile, the internal collecting tank **52** is connected to a decompression mechanism **524**. The decompression mechanism **524** decompresses the inside of the internal collecting tank **52**, to regulate a pressure of the internal chamber **520** of the internal collecting tank **52** to a negative pressure. That is, the decompression mechanism **524** decompresses the inside of the internal collecting tank **52**, to regulate a pressure of the internal chamber **520** of the internal collecting tank **52** to a pressure lower than the atmospheric pressure. The decompression mechanism **524** includes, for example, a vacuum pump, a decompression buffer tank, a pressure regulation mechanism (regulator), and the like.
[0047] The pressurization mechanism **515** and the decompression mechanism **524** are configured such that the operations thereof can be controlled by the control unit **9**. When the pressurization mechanism **515** and the decompression mechanism **524** are driven, there is generated a pressure difference between the internal chamber **510** of the supply tank **51** and the internal chamber **520** of the internal collecting tank **52**. This allows ink stored in the supply tank **51** to be supplied to the discharge heads **35**, and further allows ink remaining in the discharge heads **35** to be collected into the internal collecting tank **52**. Note that the ink remaining in the discharge heads **35** is ink being undischarged from the discharge heads **35** and left in the discharge heads **35**.
[0048] The feedback pipe **65** is a pipe connecting the internal chamber **520** of the internal collecting tank **52** and the internal chamber **510** of the supply tank **51** such that the internal chambers can communicate with each other. In other words, the feedback pipe **65** connects the internal collecting tank **52** and the supply tank **51**. As shown in FIG. **2**, an internal passage of the feedback pipe **65** has an upstream end that is connected so as to communicate with the internal chamber **520** of the internal collecting tank **52**. Further, the internal passage of the feedback pipe **65** has a downstream end that is connected so as to communicate with the internal chamber **510** of the supply tank **51**.
[0049] With the above-described configuration, there is formed an ink circulation path **41** that extends from the supply tank **51**, extends through the supply-side manifold **61**, the supply-side narrow pipes **62**, the internal tanks **82** of the discharge heads **35**, the collecting-side narrow pipes **63**, the collecting-side manifold **64**, the internal collecting tank **52**, and the feedback pipe **65**, and returns back to the supply tank **51**. In other words, the ink circulation path **41** includes the supply tank **51**, the discharge heads **35**, the internal collecting tank **52**, and the feedback pipe **65**.
[0050] Further, in the feedback pipe **65**, the circulation pump **71**, the feedback-side on-off valve **75**, the first heater **76**, the second heater **77**, the switch valve **78**, the first temperature sensor **84**, the second temperature sensor **85**, the third temperature sensor **86**, the filter **87**, and the deaeration unit **88** are interposed. The switch valve **78** provided in the ink circulation path **41** is further connected to the drainage pipe **66** as later described in detail.
[0051] The circulation pump **71** is a device configured to perform a pumping operation of delivering ink from the internal collecting tank **52** to the supply tank **51** via the feedback pipe **65**. The circulation pump **71** generates ink flow from the internal collecting tank **52** to the supply tank **51** in the internal passage of the feedback pipe **65** in response to an operation signal from the control unit **9**. For the circulation pump **71** of the present preferred embodiment, for example, a pump in which foreign matters such as dust are unlikely to be generated during driving, such as a diaphragm pump, is used. Further, the circulation pump **71** applies a pressure to ink in the circulation pump **71** by reciprocation of a piston therein. Then, the circulation pump **71** discharges ink through an outlet communicating with the internal passage of the feedback pipe **65**. Moreover,

the circulation pump **71** is electrically connected to the control unit **9**. The circulation pump **71** outputs data regarding a driving condition to the control unit **9**.

[0052] The feedback-side on-off valve **75** is interposed on the downstream side of the circulation pump **71** and on the upstream side of the first heater **76**, the second heater **77**, and the switch valve **78**, in the feedback pipe **65**. In other words, the feedback-side on-off valve **75** is interposed between the circulation pump **71** and the first and second heaters **76** and **77** in the feedback pipe **65**. The feedback-side on-off valve **75** corresponds to an “on-off valve” of the present invention. For the feedback-side on-off valve **75**, for example, a solenoid valve that is opened and closed under the control of the control unit **9** is used. Further, for the feedback-side on-off valve **75** of the present preferred embodiment, a solenoid valve of a normally closed type is used. Thus, with no power supplied, the feedback-side on-off valve **75** is closed. Alternatively, for the feedback-side on-off valve **75**, an on-off valve that is manually opened and closed may be used.

[0053] While the feedback-side on-off valve **75** is closed, the internal passage of the feedback pipe **65** is blocked from communicating. That is, while the feedback-side on-off valve **75** is closed, ink flow from the internal collecting tank **52** to the supply tank **51** and backflow of ink from the switch valve **78** to the circulation pump **71** are prevented. Meanwhile, while the feedback-side on-off valve **75** is opened, the internal passage of the feedback pipe **65** is allowed to communicate. That is, the feedback-side on-off valve **75** allows or interrupts flow of ink from the internal collecting tank **52** to the supply tank **51**.

[0054] The first heater **76** is a device configured to heat ink delivered through the internal passage of the feedback pipe **65**. The first heater **76** is positioned between the internal collecting tank **52** and the switch valve **78** in the feedback pipe **65**. The first heater **76** includes a heating element formed of a carbon heater or the like and is connected to a power supply via an ON/OFF circuit not shown. Then, the first heater **76** can heat ink by generating heat in an ON state in which power is turned on. Further, as later described in detail, the first heater **76** is controlled such that, for example, a temperature of ink passing through the first heater **76** becomes equal to a first target temperature T**1**° C. higher than a room temperature by switching between an ON state and an OFF state.

[0055] The second heater **77** is a device configured to heat ink delivered through the internal passage of the feedback pipe **65**. The second heater **77** is positioned between the first heater **76** and the switch valve **78** in the feedback pipe **65**. That is, the second heater **77** is positioned on the downstream side of the first heater **76** in the circulation path **41**. In other words, in the present preferred embodiment, the first heater **76** and the second heater **77** that are interposed in the feedback pipe **65** while being arranged adjacently to each other along the circulation path **41**, are provided. The second heater **77** includes a heating element formed of a carbon heater or the like and is connected to the power supply via the ON/OFF circuit not shown. Then, the second heater **77** can heat ink by generating heat in an ON state in which power is turned on. Further, as later described in detail, the second heater **77** is controlled such that, for example, a temperature of ink passing through the second heater **77** becomes equal to a second target temperature T**2**° C. higher than the first target temperature T**1**° C. by switching between an ON state and an OFF state. Note that the second target temperature T**2**° C. is, for example, approximately 5° C. higher than the first target temperature T**1**° C. Hereinafter, a ratio of a time period for which each of the first and second heaters **76** and **77** is in an ON state, to a unit time period, is referred to as a “duty value”. The unit time period is, for example, one second. A duty value of the second heater **77** is greater than a duty value of the first heater **76**.

[0056] However, in the present invention, the number of heaters interposed between the internal collecting tank **52** and the switch valve **78** in the feedback pipe **65** may be one or three or larger. That is, any heater that is interposed between the circulation pump **71** and the supply tank **51** in the feedback pipe **65** and heats ink flowing from the internal collecting tank **52** to the supply tank **51** can be used.

[0057] The first temperature sensor **84** is interposed on the downstream side of the feedback-side on-off valve **75** with respect to an ink delivery direction and on the upstream side of the first heater **76** with respect to the ink delivery direction in the feedback pipe **65**. The first temperature sensor **84** detects a temperature of ink flowing into the first heater **76**. Further, the first temperature sensor **84** is electrically connected to the control unit **9**. The first temperature sensor **84** outputs data regarding a result of detection of an ink temperature, to the control unit **9**.
[0058] The second temperature sensor **85** is interposed on the downstream side of the first heater **76** with respect to the ink delivery direction and on the upstream side of the second heater **77** with respect to the ink delivery direction in the feedback pipe **65**. The second temperature sensor **85** detects a temperature of ink flowing out of the first heater **76**. Further, the second temperature sensor **85** is electrically connected to the control unit **9**. The second temperature sensor **85** outputs data regarding a result of detection of an ink temperature, to the control unit **9**.
[0059] The third temperature sensor **86** is interposed on the downstream side of the second heater **77** with respect to the ink delivery direction and on the upstream side of the switch valve **78** with respect to the ink delivery direction in the feedback pipe **65**. The third temperature sensor **86** detects a temperature of ink flowing out of the second heater **77**. Further, the third temperature sensor **86** is electrically connected to the control unit **9**. The third temperature sensor **86** outputs data regarding a result of detection of an ink temperature, to the control unit **9**.
[0060] Thus, the ink supply unit **4** of the present preferred embodiment includes the temperature sensors **85** and **86** that are interposed between the first and second heaters **76** and **77** and the switch valve **78** in the feedback pipe **65** and detect a temperature of ink in the feedback pipe **65**. Note that the first temperature sensor **84**, the second temperature sensor **85**, and the third temperature sensor **86** described above are not necessarily required to be provided.
[0061] The switch valve **78** is interposed on the downstream side of the first and second heaters **76** and **77** and the temperature sensors **84**, **85**, and **86** and on the upstream side of the filter **87** in the feedback pipe **65**. In other words, the switch valve **78** is interposed between the first and second heaters **76** and **77** and the supply tank **51** in the feedback pipe **65**. The switch valve **78** is a three-way valve that switches the ink circulation path **41** in the internal passage of the feedback pipe **65** to an ink disposal path with intermediation of the drainage pipe **66**. Meanwhile, for the switch valve **78**, for example, a solenoid valve that performs a switching operation under the control of the control unit **9** is used. In the switch valve **78**, a solenoid (not shown) for switching a destination of ink flow is incorporated. Alternatively, for the switch valve **78**, a switch valve that manually performs a switching operation may be used. The switch valve **78** includes a flow inlet **781**, a first flow outlet **782**, and a second flow outlet **783**.
[0062] The flow inlet **781** communicates with the first and second heaters **76** and **77** and the temperature sensors **84**, **85**, and **86**. In other words, the flow inlet **781** communicates with an upstream-side portion of the circulation path **41**. The first flow outlet **782** communicates with the filter **87**. In other words, the first flow outlet **782** communicates with a downstream-side portion of the circulation path **41**. The second flow outlet **783** communicates with an internal passage of the drainage pipe **66**. Normally, the second flow outlet **783** is closed, and the flow inlet **781** and the first flow outlet **782** communicate with each other. That is, when power is supplied to the switch valve **78** and the solenoid is an ON state (energized), a destination of ink flow from the switch valve **78** is the first flow outlet **782**. As a result, the internal passage of the feedback pipe **65** is allowed to communicate. Meanwhile, when the first flow outlet **782** is closed and the flow inlet **781** and the second flow outlet **783** communicate with each other, ink flowing from the flow inlet **781** is directed to the disposal path with intermediation of the drainage pipe **66**, to be drained to the outside of the inkjet printing apparatus **1**. That is, when no power is supplied to the switch valve **78** and the solenoid is in an OFF state (demagnetized), the destination of ink flow from the switch valve **78** is the second flow outlet **783**. In other words, the switch valve **78** can perform a switching operation between a state in which the first and second heaters **76** and **77** communicate with the

supply tank **51** and a state in which the first and second heaters **76** and **77** are blocked from communicating with the supply tank **51** and an inner space of the circulation path **41** communicates with the external collecting tank **53** described later.

[0063] When the feedback-side on-off valve **75** is closed and the flow inlet **781** and the second flow outlet **783** communicate with each other at the switch valve **78** under the control of the control unit **9** described later, a section from the feedback-side on-off valve **75** to the switch valve **78** in the circulation path **41** is partitioned off from the other sections. Hereinafter, the section from the feedback-side on-off valve **75** to the switch valve **78** in the circulation path **41** is referred to as an "ink separation region **79**". The ink separation region **79** is set in a partial region including the first and second heaters **76** and **77** in the feedback pipe **65**. The ink separation region **79** is a region where ink in the circulation path **41** is separated from ink in the other regions in the feedback pipe **65** when the feedback-side on-off valve **75** is closed, to stop ink circulation in the circulation path **41**, and the flow inlet **781** and the second flow outlet **783** communicate with each other at the switch valve **78**. At that time, in the ink separation region **79**, ink flow to/from the other regions in the feedback pipe **65** is restricted. Note that, even though the destination of ink flow from the switch valve **78** is the first flow outlet **782**, because of flow resistance of the switch valve **78**, ink flow between the ink separation region **79** and a region on the downstream side thereof is restricted. As a result, also in this case, the ink separation region **79** is partitioned off from the other regions in the feedback pipe **65**.

[0064] The external collecting tank **53** is a container in which ink to be drained to the outside of the circulation path **41** is collected. The external collecting tank **53** is provided outside the circulation path **41** through which ink is circulated between the supply tank **51** and the internal collecting tank **52**. In the external collecting tank **53**, an internal chamber **530** in which ink can be stored is provided.

[0065] The drainage pipe **66** is a pipe that connects the internal chamber **530** of the external collecting tank **53** and the second flow outlet **783** of the switch valve **78** such that they can communicate with each other. The internal passage of the drainage pipe **66** is connected so as to communicate with the second flow outlet **783** of the switch valve **78** at an upstream end thereof. Meanwhile, the internal passage of the drainage pipe **66** is connected so as to communicate with the internal chamber **530** of the external collecting tank **53** at a downstream end thereof. By operating the switch valve **78** so as to cause the inner space of the circulation path **41** and the external collecting tank **53** to communicate with each other as described above, it is possible to deliver ink to the internal chamber **530** of the external collecting tank **53** via the internal passage of the feedback pipe **65**, the inner space of the switch valve **78**, and the internal passage of the drainage pipe **66**.

[0066] As described above, in the present preferred embodiment, the feedback-side on-off valve **75** and the switch valve **78** are provided in the ink separation region **79**. Further, when the feedback-side on-off valve **75** is closed and the destination of ink flow from the switch valve **78** is switched to the second flow outlet **783** under the control of the control unit **9**, the ink separation region **79** and the external collecting tank **53** communicate with each other. Then, ink in the ink separation region **79** is separated from ink in the other regions in the feedback pipe **65**.

[0067] The filter **87** is interposed on the downstream side of the switch valve **78** and on the upstream side of the deaeration unit **88** in the feedback pipe **65**. The filter **87** filters ink delivered through the internal passage of the feedback pipe **65**, to remove foreign matters included in the ink.

[0068] The deaeration unit **88** is interposed on the downstream side of the filter **87** and on the upstream side of the supply tank **51** in the feedback pipe **65**. The deaeration unit **88** of the present preferred embodiment is a so-called hollow-fiber membrane deaeration module. The deaeration unit **88** removes bubbles in ink delivered through the internal passage of the feedback pipe **65**.

[0069] Next, the control unit **9** is described. The control unit **9** is an information processing device configured to control each component of the inkjet printing apparatus **1**. FIG. **3** is a block diagram showing connection between the control unit **9** and each component of the inkjet printing apparatus **1**. As conceptually shown in FIG. **3**, the control unit **9** includes a processor **91** such as a CPU, a memory **92** such as a RAM, and a storage unit **93** such as a hard disk drive. In the storage unit **93**, a computer program **9**P is stored, which is for performing a printing operation while conveying the continuous paper **10**, supplying ink to the discharge heads **35** while circulating a part of ink, and draining ink to the outside of the circulation path **41**.

[0070] Further, as shown in FIG. **3**, the control unit **9** is connected to the conveyor unit **2** and the four discharge heads **35** of the printing unit **3**, and is further connected to the circulation pump **71**, the five supply-side on-off valves **73**, the five head outlet-side on-off valves **74**, the feedback-side on-off valve **75**, the first heater **76**, the second heater **77**, the switch valve **78**, the respective temperature sensors **84**, **85**, and **86**, the pressurization mechanism **515**, and the decompression mechanism **524** of each of the four ink supply units **4** of the printing unit **3** such that the control unit **9** can conduct communication to/from the above-described components. The control unit **9** controls operations of those components in accordance with the computer program **9**P.

[0071] That is, the control unit **9** can control each of the conveyor unit **2**, the four discharge heads **35** of the printing unit **3**, and the circulation pump **71**, the five supply-side on-off valves **73**, the five head outlet-side on-off valves **74**, the feedback-side on-off valve **75**, the first heater **76**, the second heater **77**, the switch valve **78**, the respective temperature sensors **84**, **85**, and **86**, the pressurization mechanism **515**, and the decompression mechanism **524** of each of the four ink supply units **4** of the printing unit **3**. By control over operations of those components in the control unit **9**, a process of conveying the continuous paper **10** and performing printing thereon proceeds, ink is supplied to the internal tank **82** of each discharge head **35** while a part thereof is circulated, and further, ink is drained to the outside of the circulation path **41** under a predetermined condition. Moreover, as described above, the circulation pump **71** outputs data regarding a driving condition of the circulation pump **71**, to the control unit **9**. The control unit **9** can detect presence or absence of ink circulation in the circulation path **41** on the basis of the data received from the circulation pump **71**.

<2. Conveyance of Continuous Paper, Printing, Ink Circulation, and Ink Drainage to Outside>

[0072] Next, description is given about a procedure for conveyance of the continuous paper **10**, printing on the continuous paper **10**, ink circulation, and ink drainage to the outside that are performed in the inkjet printing apparatus **1**. FIG. **4** is a flowchart showing a flow including conveyance of the continuous paper **10**, printing on the continuous paper **10**, ink circulation, and ink drainage ink to the outside.

[0073] First, in performing conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation, the control unit **9** causes the conveyor unit **2** to operate, to convey the continuous paper **10** along the predetermined conveying path in the conveying direction extending along the length direction. Further, the control unit **9** controls the plurality of nozzles **83** of each of the four discharge heads **35** while conveying the continuous paper **10**, to discharge ink droplets onto the surface of the continuous paper **10**. Thus, an image is recorded on the surface of the continuous paper **10** (step S**1**).

[0074] Meanwhile, as advance preparation for conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation, a sufficient amount of ink is stored in the internal chamber **510** of the supply tank **51**. Further, the control unit **9** opens the five supply-side on-off valves **73**, the five head outlet-side on-off valves **74**, and the feedback-side on-off valve **75**. Moreover, the control unit **9** closes the second flow outlet **783** of the switch valve **78**, to cause the flow inlet **781** and the first flow outlet **782** to communicate with each other.

[0075] Then, the control unit **9** drives the circulation pump **71**, the first heater **76**, the second heater **77**, the pressurization mechanism **515**, and the decompression mechanism **524** of each of the four ink supply units **4**. Specifically, the control unit **9** drives the circulation pump **71**, to circulate ink through the ink circulation path **41**, and at the same time, drives the pressurization mechanism **515** and the decompression mechanism **524**. As a result, ink is supplied to the internal tank **82** of each discharge head **35**. Further, the power of the first temperature sensor **84**, the second temperature sensor **85**, and the third temperature sensor **86** is turned on, and these sensors start measuring.

[0076] More specifically, the pressurization mechanism **515** and the decompression mechanism **524** are driven, so that a pressure difference is generated between the internal chamber **510** of the supply tank **51** and the internal chamber **520** of the internal collecting tank **52**. As a result, ink stored in the supply tank **51** can be supplied to the discharge heads **35**, and further, ink remaining in the discharge heads **35** can be collected into the internal collecting tank **52**. Note that the ink remaining in the discharge heads **35** is ink being undischarged from the discharge heads **35** and left in the discharge heads **35**. Further, when the circulation pump **71** is driven, there is generated ink flow from the internal collecting tank **52** to the supply tank **51** in the internal passage of the feedback pipe **65**.

[0077] Moreover, the first and second heaters **76** and **77** are driven, and thus ink delivered through the internal passage of the feedback pipe **65** can be delivered to the supply tank **51** after being heated. More specifically, the control unit **9** first performs control in which a temperature of ink passing through the first heater **76** becomes equal to, for example, the first target temperature T**1**° C. by switching the first heater **76** between an ON state and an OFF state. As described above, the second temperature sensor **85** detects a temperature of ink flowing out of the first heater **76** and outputs data regarding a result of the detection, to the control unit **9**. The control unit **9** adjusts a driving amount of the first heater **76** in accordance with the result of the detection of the ink temperature received from the second temperature sensor **85**.

[0078] FIG. **5** is a view showing examples of a temperature of ink passing through each of the first and second heaters **76**, **77** and a ratio of a time period of an ON state to a unit time period in driving the first and second heaters **76** and **77**, plotted against the temperature of ink. The unit time period is, for example, one second. In other words, FIG. **5** is a view showing examples of a temperature of ink passing through each of the first and second heaters **76** and **77** and a duty value in driving each of the first and second heaters **76** and **77**, plotted against the temperature of ink. As shown in FIG. **5**, when a temperature of ink passing through the first heater **76** is lower than the "first target temperature T**1**−2.5° C.", the control unit **9** drives the first heater **76** at a duty value of 100%. When the temperature of ink passing through the first heater **76** becomes equal to or higher than the "first target temperature T**1**−2.5° C.", the control unit **9** gradually decreases the duty value. Then, when the temperature of ink becomes equal to or higher than the "first target temperature T**1**−2° C.", the control unit **9** drives the first heater **76** at a duty value of 50% or smaller. Further, when the temperature of ink becomes equal to or higher than the "first target temperature T**1**° C.", the control unit **9** drives the first heater **76** at a duty value of 0%. In other words, when the temperature of ink becomes equal to or higher than the "first target temperature T**1**° C.", the control unit **9** stops driving the first heater **76**. By the above-described control, the temperature of ink passing through the first heater **76** can be adjusted to the "first target temperature T**1**° C.".

[0079] Meanwhile, the control unit **9** performs control in which a temperature of ink passing through the second heater **77** becomes equal to, for example, the second target temperature T**2**° C. by switching the second heater **77** between an ON state and an OFF state. As described above, the third temperature sensor **86** detects a temperature of ink flowing out of the second heater **77**, and outputs data regarding a result of the detection, to the control unit **9**. The control unit **9** adjusts a driving amount of the second heater **77** in accordance with the result of the detection of the ink temperature received from the third temperature sensor **86**. Here, as described above, a duty value of the second heater **77** is greater than a duty value of the first heater **76**.

[0080] As shown in FIG. **5**, when the temperature of ink passing through the second heater **77** is lower than the "first target temperature T**1**+3° C." (corresponding to the second target temperature T**2**−2° C.), the control unit **9** drives the second heater **77** at a duty value of 100%. When the temperature of ink passing through the second heater **77** becomes equal to or higher than the "first target temperature T**1**+3° C.” (corresponding to the second target temperature T**2**−2° C.), the control unit **9** gradually decreases the duty value. Then, when the temperature of ink becomes equal to or higher than the “first target temperature T**1**+4° C.” (corresponding to the second target temperature T**2**−1° C.), the control unit **9** drives the second heater **77** at a duty value of 50% or smaller. Further, when the temperature of ink becomes equal to or higher than the “first target temperature T**1**+5° C.” (corresponding to the second target temperature T**2**° C.), the control unit **9** drives the second heater **77** at a duty value of 0%. In other words, when the temperature of ink becomes equal to or higher than the “first target temperature T**1**+5° C.” (corresponding to the second target temperature T**2**° C.), the control unit **9** stops driving the second heater **77**. By the above-described control, the temperature of ink passing through the second heater **77** can be adjusted to the “second target temperature T**2**° C.”. The adjustment of the temperature of ink passing through the second heater **77** to the second target temperature T**2**° C. allows the discharge heads **35** to satisfactorily discharge ink afterward.

[0081] Meanwhile, as described above, the first temperature sensor **84** detects a temperature of ink flowing into the first heater **76**, and outputs data regarding a result of the detection, to the control unit **9**. The control unit **9** can find at what temperature ink has originally flowed into the first heater **76**, by referring to the data regarding the result of the detection provided from the first temperature sensor **84**. This serves as a measure in determination as to whether the first heater **76** and the second heater **77** normally operate.

[0082] Further, ink flowing through the internal passage of the feedback pipe **65** is caused to pass through the filter **87**, and thus, ink from which small impurities or the like remaining therein have been removed can be delivered to the supply tank **51**. Moreover, ink flowing through the internal passage of the feedback pipe **65** is caused to pass through the deaeration unit **88**, and thus ink having been deaerated can be delivered to the supply tank **51**.

[0083] Here, suppose that a power failure occurs and power supply to the inkjet printing apparatus **1** is interrupted, for example. In this case, also power supply to the circulation pump **71** is interrupted, so that the circulation pump **71** stops operating. In another supposed case, when an operator detects ink leakage from the ink circulation path **41**, an excessive decrease of the liquid level of ink stored in the internal chamber **510** of the supply tank **51**, or overflow of ink stored in the internal chamber **530** of the external collecting tank **53**, the operator himself stops driving the circulation pump **71**. In those cases, ink circulation in the circulation path **41** is stopped, and thus ink stays in each of the first and second heaters **76** and **77**.

[0084] In this regard, as described above, each of the first and second heaters **76** and **77** includes a heating element formed of a carbon heater or the like. The heating element is kept at a high temperature even after interruption of power supply to the first and second heaters **76** and **77**. Hence, to leave ink circulation stopped for a certain time period would result in excessively heating the ink staying in the first and second heaters **76** and **77**, so that the ink might possibly be changed in quality or deteriorated. Further, to use ink having been changed in quality or deteriorated afterward would cause a fear that ink might not be satisfactorily discharged from the discharge heads **35** or a desired color gamut might not be obtained.

[0085] Thus, the control unit **9** first detects whether the circulation pump **71** has stopped operating (step S**2**). A state in which the circulation pump **71** has stopped operating includes a case in which a power failure occurs and the circulation pump **71** automatically stops operating due to interruption of power supply to the circulation pump **71**, and a case in which an operator himself stops driving the circulation pump **71**. The control unit **9** continues conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation until it is detected that the circulation pump **71** has stopped operating (step S**2**: NO). Meanwhile, when it is detected that the circulation pump **71** has stopped operating (step S**2**: YES), the control unit **9** performs an ink separation step S**21** described below.

[0086] The feedback-side on-off valve **75** is a solenoid valve of a normally closed type as described above, and thus, is automatically closed when power supply to the feedback-side on-off valve **75** is interrupted due to a power failure or the like. Alternatively, the control unit **9** may transmit a control signal for interrupting ink flow to the feedback-side on-off valve **75** when the control unit **9** detects a stop of ink circulation in the circulation path **41**. Further alternatively, an operator may manually close the feedback-side on-off valve **75**. Thus, when ink circulation in the circulation path **41** stops due to a power failure or the like, the ink separation region **79** is partitioned off from the other regions in the feedback pipe **65** (step S**21**).

[0087] Further, as described above, when the solenoid incorporated in the switch valve **78** is in a demagnetized (OFF) state, a destination of ink flow is switched to the second flow outlet **783**. Thus, when power supply to the switch valve **78** stops due to a power failure or the like, a destination of ink flow is automatically switched to the second flow outlet **783**. Alternatively, the control unit **9** may transmit a control signal for switching a destination of ink flow to the second flow outlet **783**, to the switch valve **78** when the control unit **9** detects a stop of ink circulation in the circulation path **41**. Further alternatively, an operator may manually switch a destination of ink flow of the switch valve **78** to the second flow outlet **783**. In the above-described manner, the ink separation step S**21** is performed.

[0088] After performing the ink separation step S**21**, the control unit **9** recovers from a power failure and eliminates ink leakage or the like, and after then performs a post-recovery process (step S**3**) when performing conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation again.

[0089] FIG. **6** is a flowchart showing a procedure for performing the post-recovery process. As described in detail below, for the post-recovery process, the control unit **9** can selectively perform a “normal resumption mode” and a “post-drainage resumption mode”. That is, after the ink separation step S**21** is performed and recovery from a power failure is achieved or ink leakage or the like is eliminated, the control unit **9** selectively performs the “normal resumption mode” and the “post-drainage resumption mode”. Meanwhile, the control unit **9** can determine whether recovery from a power failure has been achieved or whether ink leakage or the like has been eliminated, according to presence or absence of resumption of power supply to the inkjet printing apparatus **1**, or from an input operation or the like by an operator.

[0090] As shown in FIG. **6**, before resuming ink circulation in the circulation path **41**, the control unit **9** first determines whether a temperature of ink in the ink separation region **79** is lower than a predetermined threshold value, or equal to or higher than the predetermined threshold value (step S**31**). To this end, the control unit **9** refers to each of a duty value of the first heater **76** and a duty value of the second heater **77** for a first predetermined time period before it is detected that ink circulation in the circulation path **41** has stopped, in making the foregoing determination. For example, the control unit **9** refers to each of a duty value of the first heater **76** and a duty value of the second heater **77** in one second long being prior to the point at which it is detected that ink circulation in the circulation path **41** has stopped. Note that the duty value varies in some cases, as described above. Hence, the control unit **9** may set a time period of approximately a few minutes as the first predetermined time period and refer to integral values of duty values of the respective first and second heaters **76** and **77** for the set time period.

[0091] When the duty value for the first predetermined time period before it is detected that ink circulation has stopped is large, the control unit **9** determines that the respective heating elements have continued generating much heat in the first and second heaters **76** and **77** also during a stop of ink circulation. In this case, the control unit **9** determines that ink staying in the first and second heaters **76** and **77** has been excessively heated and is highly likely to have been changed in quality or deteriorated. Then, the control unit **9** determines whether a temperature of ink in the ink separation region **79** is lower than, or equal to or higher than, the threshold value on the basis of a result of referring to the duty values.

[0092] More specifically, after detecting that ink circulation in the circulation path **41** has stopped, the control unit **9** determines whether a temperature of ink in the first heater **76** is lower than, or equal to or higher than, a first predetermined threshold value before resuming ink circulation in the circulation path **41**. Meanwhile, after detecting that the ink circulation in the circulation path **41** has stopped, the control unit **9** determines whether a temperature of ink in the second heater **77** is lower than, or equal to or higher than, a second predetermined threshold value before resuming ink circulation in the circulation path **41**. In this manner, in the present preferred embodiment, ink can be drained as later described in accordance with a result of determination as to whether an ink temperature is lower than, or equal to or higher than, the predetermined threshold value with consideration of heating values of the first and second heaters **76** and **77**.

[0093] Alternatively, the control unit **9** may determine whether the temperature of ink in the ink separation region **79** is lower than, or equal to or higher than, the predetermined threshold value on the basis of a result of detection provided from the second temperature sensor **85** placed on the downstream side of the first heater **76** and a result of detection provided from the third temperature sensor **86** placed on the downstream side of the second heater **77**. Then, in ink drainage described later, when it is determined that ink is at a sufficiently low temperature on the basis of the results of detections provided from the temperature sensors **85** and **86**, the control unit **9** may stop draining ink to the outside at that time point.

[0094] Then, when it is determined that the temperature of ink in the ink separation region **79** is lower than the predetermined threshold value (step S**31**: YES), the control unit **9** performs the “normal resumption mode” (step S**32**). More specifically, when it is determined that the temperature of ink in the first heater **76** is lower than the first predetermined threshold value and the temperature of ink in the second heater **77** is lower than the second predetermined threshold value, the control unit **9** performs the “normal resumption mode”. That is, the control unit **9** performs the “normal resumption mode” in accordance with the temperature of ink in the ink separation region **79**. In this case, for the “normal resumption mode”, the control unit **9** causes the ink separation region **79** and the other regions in the feedback pipe **65** to communicate with each other immediately. Specifically, the control unit **9** closes the second flow outlet **783** of the switch valve **78** and causes the flow inlet **781** and the first flow outlet **782** to communicate with each other, and thus the first and second heaters **76** and **77** communicate with the supply tank **51**. Further, the control unit **9** opens the feedback-side on-off valve **75**. Subsequently, the control unit **9** drives the circulation pump **71** to resume ink circulation, and performs conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation.

[0095] Meanwhile, when it is determined that the temperature of ink in the ink separation region **79** is equal to or higher than the predetermined threshold value (step S**31**: NO), the control unit **9** performs the “post-drainage resumption mode” (step S**33**). For example, when it is determined that the temperature of ink in the first heater **76** is lower than the first predetermined threshold value and the temperature of ink in the second heater **77** is equal to or higher than the second predetermined threshold value, the control unit **9** performs the “post-drainage resumption mode”. That is, the control unit **9** performs the “post-drainage resumption mode” in accordance with the temperature of ink in the ink separation region **79**. In this case, for the “post-drainage resumption mode”, the control unit **9** closes the first flow outlet **782** of the switch valve **78**, to cause the flow inlet **781** and the second flow outlet **783** to communicate with each other. As a result, the inner space of the circulation path **41** and the external collecting tank **53** communicate with each other. Further, the control unit **9** opens the feedback-side on-off valve **75**. Then, the control unit **9** drives the circulation pump **71** for a “specific time period” and drains ink in the ink separation region **79** to the external collecting tank **53**. After that, the control unit **9** causes the ink separation region **79** and the other regions in the feedback pipe **65** to communicate with each other again, as described later.

[0096] Here, the “specific time period” is, for example, a value obtained by division of a sum of a volume of a section from the switch valve **78** to the second heater **77** in the internal passage of the feedback pipe **65** and a volume of the inner space of the second heater **77**, by a pumping amount of the circulation pump **71** per unit time period. In other words, the "specific time period" is a value obtained by division of a specific capacity calculated from an ink-passable capacity of the second heater **77**, by a pumping amount of the circulation pump **71** per unit time period. Thus, ink that is likely to have been deteriorated due to overheating by the second heater **77** can be drained to the outside of the circulation path **41**. Further, it is possible to narrow ink to be drained at that time, down to ink that is likely to have been deteriorated due to overheating by the second heater **77**.

[0097] In this regard, an amount of ink to be drained to the external collecting tank **53** may be all of ink in the ink separation region **79**. Alternatively, an amount of ink to be drained to the external collecting tank **53** may be an amount that is further reduced by referring to the temperature of ink in the first heater **76** and the temperature of ink in the second heater **77**. That is, the control unit **9** selects the "post-drainage resumption mode" in accordance with the temperature of ink in the ink separation region **79**, opens the feedback-side on-off valve **75**, and causes the ink separation region **79** and the external collecting tank **53** to communicate with each other at the switch valve **78**. Then, the control unit **9** drives the circulation pump **71** and drains at least a part of ink in the ink separation region **79** to the external collecting tank **53**. After that, the control unit **9** causes the ink separation region **79** and the other regions in the feedback pipe **65** to communicate with each other again as described later.

[0098] Further, when it is determined that the temperature of ink in the first heater **76** is equal to or higher than the first predetermined threshold value and the temperature of ink in the second heater **77** is equal to or higher than the second predetermined threshold value, the control unit **9** performs the "post-drainage resumption mode" in the same manner as described above. In this case, the control unit **9** operates the switch valve **78** in the same manner as described above, opens the feedback-side on-off valve **75**, and drives the circulation pump **71** for a "specific time period". In this case, the "specific time period" is, for example, a value obtained by division of a sum of a volume of a section from the switch valve **78** to the first heater **76** in the internal passage of the feedback pipe **65**, a volume of the inner space of the second heater **77**, and a volume of the inner space of the first heater **76**, by a pumping amount of the circulation pump **71** per unit time period.

[0099] Specifically, when it is determined that the temperature of ink in the ink separation region **79** is equal to or higher than the predetermined threshold value, the control unit **9** causes the ink separation region **79** and the external collecting tank **53** to communicate with each other at the switch valve **78**. Further, the control unit **9** opens the feedback-side on-off valve **75**. Then, the circulation pump **71** is driven for the "specific time period", and ink in the ink separation region **79** is drained to the external collecting tank **53**. Thus, in a case in which ink circulation stops, ink that is likely to have been changed in quality or deteriorated due to overheating can be drained to the outside in accordance with the temperature of the ink, in recovery afterward. Meanwhile, in draining ink to the external collecting tank **53**, all of ink in the circulation path **41** is not drained. In the present preferred embodiment, only ink that is likely to have been changed in quality or deteriorated due to overheating by the second heater **77**, or only ink that is likely to have been changed in quality or deteriorated due to overheating by both of the first heater **76** and the second heater **77**, in the ink separation region **79**, is drained to the external collecting tank **53**. This enables reduction of an amount of ink to be drained.

[0100] Meanwhile, in performing the "post-drainage resumption mode", the control unit **9** stops driving the first and second heaters **76** and **77** at least until the ink separation region **79** and the other regions in the feedback pipe **65** communicate with each other as described later. This enables further suppression of overheating of ink.

[0101] As described above, after the circulation pump **71** is driven for the "specific time period" and ink in the ink separation region **79** is drained to the external collecting tank **53**, the control unit **9** closes the second flow outlet **783** of the switch valve **78** again. Thus, the control unit **9** causes the flow inlet **781** and the first flow outlet **782** to communicate with each other, to thereby cause the first and second heaters **76** and **77** and the supply tank **51** to communicate with each other. That is, the control unit **9** causes the ink separation region **79** and the other regions in the feedback pipe **65** to communicate with each other again. Further, the control unit **9** opens the feedback-side on-off valve **75**. Then, the control unit **9** drives the circulation pump **71** to resume ink circulation, and performs conveyance of the continuous paper **10**, printing on the continuous paper, and ink circulation. Moreover, the control unit **9** drives the first and second heaters **76** and **77** again, and adjusts driving amounts of the first and second heaters **76** and **77** in accordance with results of detection of ink temperatures received form the temperature sensors **85** and **86**.

[0102] As shown in FIG. **4**, after that, the control unit **9** determines whether to end conveyance of the continuous paper **10** and printing (step S**4**). The control unit **9** continues conveyance of the continuous paper **10** and printing when there remains image data to be printed (step S**4**: NO). After a while, when there is no image data to be printed (step S**4**: YES), the control unit **9** stops operations of each component and ends conveyance of the continuous paper **10**, printing on the continuous paper **10**, and ink circulation.

<3. Modifications>

[0103] Hereinabove, the preferred embodiment of the present invention has been described, but the present invention is not limited to the above-described preferred embodiment.

[0104] In the above-described preferred embodiment, in draining ink in the ink separation region **79** to the external collecting tank **53** in the “post-drainage resumption mode”, the feedback-side on-off valve **75** is opened and the circulation pump **71** is driven. Alternatively, ink in the inner space of the circulation path **41** may be drained to the external collecting tank **53** by pressurization of a side where the circulation path **41** is provided and decompression of a side where the external collecting tank **53** is provided, using a pressurization mechanism and a decompression mechanism that are additionally provided, without opening the feedback-side on-off valve **75** or driving the circulation pump **71**.

[0105] Further, the respective elements described in the above-described preferred embodiment and the modification may be appropriately combined unless contradiction occurs.

[0106] While the invention has been shown and described in detail, the foregoing description is in all aspects illustrative and not restrictive. It is therefore understood that numerous modifications and variations can be devised without departing from the scope of the invention.

Claims

1. An inkjet printing apparatus that discharges ink onto a printing medium to perform printing, comprising: an ink circulation path including a discharge head configured to discharge ink, a supply tank in which ink supplied to the discharge head is stored, an internal collecting tank in which ink collected from the discharge head is stored, and a feedback pipe connecting the internal collecting tank and the supply tank; an external collecting tank in which ink drained to outside of the circulation path is collected; a circulation pump that is interposed in the feedback pipe and is configured to deliver ink from the internal collecting tank to the supply tank via the feedback pipe; a heater that is interposed between the circulation pump and the supply tank in the feedback pipe and is configured to heat ink flowing from the internal collecting tank to the supply tank; a control unit configured to control the discharge head, the circulation pump, and the heater, and detect presence or absence of ink circulation in the circulation path; and an ink separation region in which ink in the circulation path is separated from ink in the other regions in the feedback pipe when the ink circulation in the circulation path stops, the ink separation region being set in a partial region including the heater in the feedback pipe, wherein, the control unit selectively performs a normal resumption mode in which the ink separation region and the other regions in the feedback pipe are caused to communicate with each other immediately, and a post-drainage resumption mode in which the ink separation region and the other regions in the feedback pipe are caused to communicate with each other after at least a part of ink in the ink separation region is drained to the external collecting tank, in accordance with a temperature of the ink in the ink separation region, before resuming the ink circulation in the circulation path.

2. The inkjet printing apparatus according to claim 1, further comprising: an on-off valve that is interposed between the circulation pump and the heater in the feedback pipe and is configured to allow or interrupt ink flow from the internal collecting tank to the supply tank; and a switch valve that is interposed between the heater and the supply tank in the feedback pipe and is configured to perform switching between a state in which the heater and the supply tank communicate with each other and a state in which the heater and the supply tank are blocked from communicating with each other, and in the latter state, an inner space of the circulation path and the external collecting tank communicate with each other, wherein the on-off valve and the switch valve are provided in the ink separation region, the control unit is configured to further control the on-off valve and the switch valve, and under the control of the control unit, the on-off valve is closed, and the ink separation region and the external collecting tank communicate with each other at the switch valve, so that the ink in the ink separation region is separated from the ink in the other regions in the feedback pipe.

3. The inkjet printing apparatus according to claim 2, wherein, in the post-drainage resumption mode, under the control of the control unit, the on-off valve is opened, and the circulation pump is driven while the ink separation region and the external collecting tank communicate with each other at the switch valve, so that at least the part of the ink in the ink separation region is drained to the external collecting tank.

4. The inkjet printing apparatus according to claim 1, wherein the control unit determines whether the temperature of the ink in the ink separation region is lower than a predetermined threshold value, or is equal to or higher than the predetermined threshold value, performs the normal resumption mode when it is determined that the temperature of the ink in the ink separation region is lower than the predetermined threshold value, and performs the post-drainage resumption mode when it is determined that the temperature of the ink in the ink separation region is equal to or higher than the predetermined threshold value.

5. The inkjet printing apparatus according to claim 4, wherein the control unit determines whether the temperature of the ink in the ink separation region is lower than, or equal to or higher than, the predetermined threshold value on the basis of a duty value of the heater for a first predetermined time period before it is detected that the ink circulation in the circulation path has stopped.

6. The inkjet printing apparatus according to claim 4, further comprising a temperature sensor that is interposed between the heater and the switch valve in the feedback pipe and is configured to detect a temperature of ink in the feedback pipe, wherein the control unit determines whether the temperature of the ink in the ink separation region is lower than, or equal to or higher than, the predetermined threshold value on the basis of a result of detection provided from the temperature sensor.

7. The inkjet printing apparatus according to claim 1, wherein, in performing the post-drainage resumption mode, the control unit stops driving the heater at least until the ink separation region and the other regions in the feedback pipe communicate with each other.

8. The inkjet printing apparatus according to claim 2, further comprising a first heater and a second heater that are interposed in the feedback pipe while being arranged adjacently to each other along the circulation path, as the heater, wherein the second heater is positioned on a downstream side of the first heater in the circulation path and has a duty value greater than a duty value of the first heater, before resuming the ink circulation in the circulation path, the control unit determines whether a temperature of ink in the first heater is lower than, or equal to or higher than, a first predetermined threshold value, and whether a temperature of ink in the second heater is lower than, or equal to or higher than, a second predetermined threshold value, when it is determined that the temperature of the ink in the first heater is lower than the first predetermined threshold value and the temperature of the ink in the second heater is equal to or higher than the second predetermined threshold value, the control unit performs the post-drainage resumption mode in which the inner space of the circulation path and the external collecting tank are caused to communicate with each other at the switch vale, the circulation pump is driven for a specific time period, the ink in the ink separation region is drained to the external collecting tank, and then, the ink separation region and the other regions in the feedback pipe are caused to communicate with each other at the switch valve, and the specific time period is obtained by division of a specific capacity calculated from an ink-passable capacity of the second heater, by a pumping amount of the circulation pump per unit time period.